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数学建模国际赛

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编 号 专 用 页

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THE ANALYSIS OF EBOLA VIRUS

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Abstract:

According to the question, we study the Ebola virus in each aspect. We check the data on the Internet about Ebola virus outbreak in Africa. Compare with different country's case of illness ,rate of death and others factors which affected most serious. Finding the internal information and the characteristic of problem is not very obvious to us, in other words ,part of them is unknown to us. Then we find the different mathematical model to solve it. Using the data we collect to draw the rate curve and predict it's change in the future.

Focusing on the task#1, we find the finical of America's costs in different aspects. Predicting the economic development in drugs for cure Ebola virus. And forecast how much America will spend to deal with the crisis. Collecting the informations about virus in Africa. Connected the economy, virus and so many factors, finding the suitable country. Using gray prediction model and GM(1,1) model to build the codes in MATLAB then draw the curve of different factors.

Task#2 is refer to the Polynomial regression curve fitting and we use the Logistic regression model to solve it. We are fitting the different data them together. Through the two models we predict the probability for drugs development. At last, we analyze the result from the Ebola virus. We study at the influence factor of drugs research. Finding the main influence resource is the virus development spread, rate of death, deathrate of change. Fitting three factors into the curve of change.

Focusing on task#3, we build the "reward" policy model describe it impersonality in every aspect of "reward" policy. After that, on the basis of result of development divided offer us a new innovate "reward" policy. It excites the company to create bigger social benefit. And make "reward" policy become better.

In task#4, in line accordance of every company's scheme dealing with risks of drugs development and different costs. Through the task assignment model, lower the cost and the risks of drug development. Making a better optimal assignment to increase the effective of work.

Key words:

Gray Prediction Model, GM(1,1)Model, Logistic regression model, Generalized Linear Model, Probabilistic Models, "reward" policy model, Innovative Incentives, Tasking Matching Model

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I. Introduction

In order to indicate the origin of Ebola virus problems, the following background is worth mentioning.

1.1 The problem itself

The deadly hemorrhagic fever Ebola was first discovered in 1976, and it has haunted the public imagination for twenty years, ever since the publication of Richard Preston's "The Hot Zone." Yet, in all that time, no drug has ever been approved to treat the disease. Since December 2013, an ongoing outbreak of Ebola in West Africa has infected at least 567 people in Guinea, Sierra Leone and Liberia, including 350 who died, according to the World Health Organization. The outbreak appears to be the largest in history, surpassing the 425 cases that occurred in an Ebola outbreak in Uganda in 2000. People with Ebola are treated with only general therapies meant to support the ill patient. They might be given fluids (Ebola patients are frequently dehydrated), or treatments aimed at maintaining blood pressure and oxygen levels, and treating infections if they develop, according to the Centers for Disease Control and Prevention. (Supplies of the experimental drug administered to two American patients have already run out.) The lack of an Ebola treatment is disturbing. But, given the way drug development is funded, it's also predictable.

When pharmaceutical companies are deciding where to direct their R & D money, they naturally assess the potential market for a drug candidate. That means that they have an incentive to target diseases that affect wealthier people (above all, people in the developed world), who can afford to pay a lot. They have an incentive to make drugs that many people will take. And they have an incentive to make drugs that people will take regularly for a long time—drugs like statins.

This system does a reasonable job of getting Westerners the drugs they want (albeit often at high prices). But it also leads to enormous underinvestment in certain kinds of diseases and certain categories of drugs. Diseases that mostly affect poor people in poor countries aren't a research priority, because it's unlikely that those markets will ever provide a decent return. So diseases like malaria and tuberculosis, which together kill two million people a year, have received less attention from pharmaceutical companies than high cholesterol. Then, there's what the World Health Organization calls "neglected tropical diseases," such as Chagas disease and dengue; they affect more than a billion people and kill as many as half a million a year. One study found that of the more than fifteen hundred drugs that came to market between 1975 and 2004 just ten were targeted at these maladies. And when a disease's victims are both poor and not very numerous that's a double whammy. On both scores, a drug for Ebola looks like a bad investment: so far, the disease has appeared only in poor countries and has affected a relatively small number of people.

It's not just developing nations that the system disservices, however. In recent years, the rise of drug-resistant microbes has made the antibiotics we use less effective and has increased the risk that an infectious disease could get out of control. What people in the West need, health officials agree, is new drugs that we can keep in reserve against an outbreak that regular antibiotics can't contain. Yet, over the past thirty years, the supply of new antibiotics has slowed to a trickle. "Antibiotic resistance really has the potential to make everything about the way we live different," Kevin Outterson, a co-director of the Health Law program at

Boston University and a founding member of the C.D.C.'s working group on antimicrobial resistance, told me. "So we need to stoke the pipeline."

The trouble, again, is the business model. If a drug company did invent a powerful new antibiotic, we would n't want it to be widely prescribed, because the goal would be to delay resistance. "Public-health officials would appropriately try to limit sales of the drug as much as possible," Outtersen says: a good public-health policy; a bad investment prospect.

So how can we get the drugs we need without magically transforming the industry that develops them? The key is to reward companies for creating substantial public-health benefits. And the simplest way to do this would be to offer prizes for new drugs. Outtersen describes one scenario: "The government would make a payment or a stream of payments to the company, and in exchange the company would give up the right to sell the product." The drug company would get paid, and would avoid all the expenses of trying to push a new product (which you don't want with a last-resort antibiotic, anyway). Society would get a new drug, and public-health officials would be able to control how it was promoted and used.

Task #1: Choose the country you believe to be most critical in terms of Ebola in Africa. Build a model to approximate the expected rate of change in the number of Ebola infections for the country from 2006 to 2025, in the absence of any additional drugs. Fully explain your model and the assumptions that underlie your model. In addition, estimate the cost of US to deal with the crisis.

Task #2: Build a mathematical model to determine whether any new disease need to develop drugs during this time period in the area. If so, estimate the deadline and output of drug development process.

Task #3: Develop a "reward" policy for pharmaceutical companies to support drug R & D work. The approach that you develop should make use of public-health benefits and US Budget.

Task #4: Joint drug research conducted by multiple companies can effectively lower costs and risk. How to select some enterprises to establish cooperation is a thorny issue. Your team can help the government to make a choice?

1.2 What we faced the annoyance in problems

In task#1, we check the informations in the Internet about Ebola virus outbreak in Africa. Compare with different country's case of illness ,rate of death and others factors which affected most serious. In task#1, the internal information and the characteristic of problem is not very obvious to us, in other words ,part of them is unknown to us. Then we find the Gray Prediction Model to solve it. Using the data we collect to draw the rate curve and predict it's change in next eleven years.

Then we find the finical of America's costs in different aspects. Predicting the economic development in drugs for cure Ebola virus. And forecast how much America will spend to deal with the crisis.

1.3 Operational research

The operational research is a subject connected with applied mathematics and form of science. Using statistics, mathematical model and algorithm and other methods to find the to find the optimal solution or optimal answer. Operational research often used in the complex question in daily life and change effective of the existing system. And we use the operational research in our models.

1.4 Probability theory

Probability theory is a subject study random event and research the number 0 and 1, 1 means occurring event, 0 means not occurring event. Probability is between 0 and 1; The formula of probability theory:

$$P_{(A)} = \frac{m}{n}$$

We use probability theory in the codes in MATLAB by following.

II. The Description of the Problem

2.1 The physical truth of Democratic Republic of Congo

Surfing in the Internet we find the Democratic Republic of Congo is the country which the death rate of Ebola virus are the highest in Africa.

In the chart it show us the Ebola virus outbreak in Africa.

years	country	Case of illness	death	Rate of death
2005	Congo	12	10	83.3%
2004	Sudan	17	7	41.2%
2003	Congo	178	157	88.2%
2001-2002	Congo	59	44	74.6%
2001-2002	Gabon	65	53	81.5%
2000	Uganda	425	224	52.7%
1996	Gabon	91	66	72.5%
1995	Democratic Republic of Congo	315	254	80.6%
1994	Ivory Coast	1	0	0%
1994	Gabon	52	31	59.6%
1979	Sudan	34	22	64.7%
1977	Democratic Republic of Congo	1	1	100%
1976	Sudan	284	151	53.2%
1976	Democratic Republic of Congo	318	280	88.1%

According to the chart, we can conclude the Democratic Republic of Congo is the country which Ebola virus is most serious.

III. Models

3.1 Gray Prediction Model (we use this model into Task#1)

3.1.1 The Foundation of Gray Prediction Model

We build the curve with the Gray Prediction Model. If the internal information and the characteristic of a system is not very obvious to us, in other words, part of them is unknown to us. It called the Gray Prediction Model. The Gray Prediction Model's primary mission is searching the factor and its mathematic's connection to build the suitable model, according to the behavioral characteristics of data in gray system and make best use of the information whether it obvious or not.

3.1.2 Gray generating sequence

In gray system theory, we take all the random variable as the gray number, Dealing with the random variable meanings that we have to find the inherent laws, then we will produce the new data column.

(1)The accumulated generating

The process of sequence data accumulate in every moment we called accumulated generating. The new sequence produce by the accumulated generating it named Accumulated generating sequence.

Assume the original sequence $x^{(0)}$ is

$$x^{(0)} = [x^{(0)}(1), x^{(0)}(2), \dots, x^{(0)}(n)]$$

$$x^{(1)}(k) = \sum_{i=1}^k x^{(0)}(i) \quad (k = 1, 2, 3, \dots, n)$$

We call $x^{(1)}(k)$ as one to the power of the sequence $x^{(0)}$'s accumulated generating. The sequence $x^{(1)} = [x^{(1)}(1), x^{(1)}(2), \dots, x^{(1)}(n)]$ called after one to the power of accumulated generating sequence in $x^{(0)}$.

Similar, as the sequence following list

$$x^{(r)}(k) = \sum_{i=1}^k x^{(r-1)}(i) \quad (k = 1, 2, 3, \dots, n, \quad r \geq 1)$$

It named R to the power of accumulated generating in $x^{(0)}$.

Sequence

$$x^{(r)} = [x^{(r)}(1), x^{(r)}(2), \dots, x^{(r)}(n)]$$

called R to the power of accumulated generating sequence in $x^{(0)}$.

(2)Mean Generating

Set the original sequence

$$x^{(0)} = [x^{(0)}(1), x^{(0)}(2), \dots, x^{(0)}(k-1), x^{(0)}(k), \dots, x^{(0)}(n)]$$

we called $x^{(0)}(k-1)$ and $x^{(0)}(k)$ are adjacent values of sequence $x^{(0)}$, the former neighbor values is $x^{(0)}(k)$, the later neighbor values is $x^{(0)}(k-1)$

As usual constant $\alpha \in [0,1]$,

$$z^{(0)}(k) = \alpha x^{(0)}(k) + (1-\alpha)x^{(0)}(k-1)$$

It is adjacent values generation number that under the condition of the adjacent values of sequence $x^{(0)}$ generation coefficient.

Particularly, when the coefficient generator $\alpha = 0.5$,

$$z^{(0)}(k) = 0.5x^{(0)}(k) + 0.5x^{(0)}(k-1)$$

It is the mean number of adjacent formation.

Similarly, we can define the non adjacent values generated number:

$$z^{(0)}(k) = \alpha x^{(0)}(k+1) + (1-\alpha)x^{(0)}(k-1)$$

And

$$z^{(0)}(k) = 0.5x^{(0)}(k+1) + 0.5x^{(0)}(k-1)$$

Sequence

is named the mean generating sequence.

3.1.3 GM(1,1) Model

3.1.3.1 The Foundation of GM(1,1) Model

Let $x^{(0)}$ be a n element number sequence that $x^{(0)} = (x^{(0)}(1), x^{(0)}(2), \dots, x^{(0)}(n))$,

the get sequence AGO of $x^{(0)}$ is

$$x^{(1)} = (x^{(1)}(1), x^{(1)}(2), \dots, x^{(1)}(n))$$

$$x^{(1)}(k) = \sum_{j=1}^k x^{(0)}(j) \quad (k=1,2,3,\dots,n)$$

Then we define the gray derivative of $x^{(1)}$ is

$$d(k) = x^{(0)}(k) = x^{(1)}(k) - x^{(1)}(k-1),$$

let $z^{(1)}$ be the close to the mean sequence of the sequence $x^{(1)}$, that's mean:

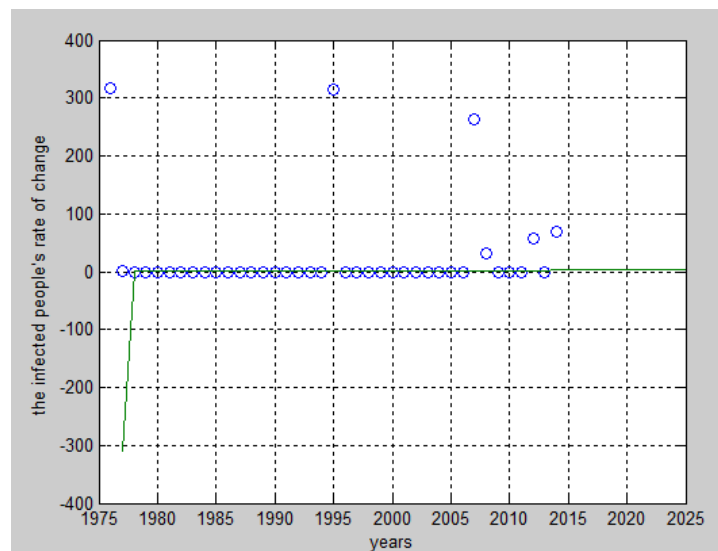
$$z^{(1)}(k) = 0.5x^{(1)}(k) + 0.5x^{(1)}(k-1), k=2,3,\dots,n$$

And

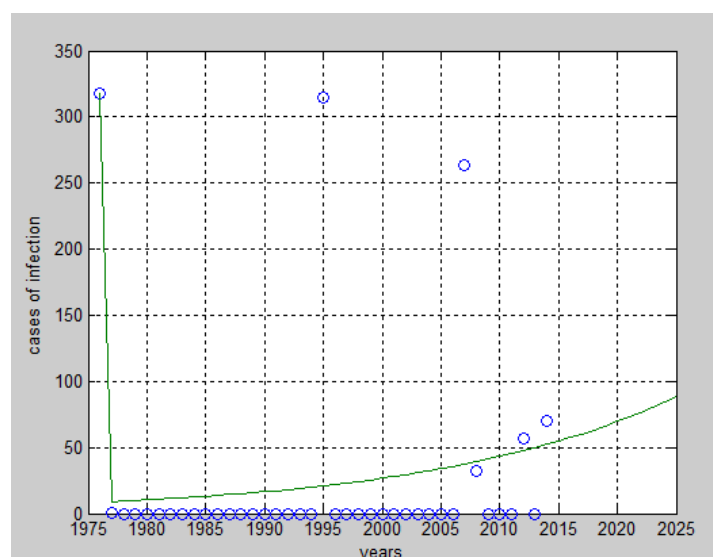

```

D=D';
E=[-C;ones(1,n-1)];
c=inv(E*E')*E*D;
c=c';
a=c(1);b=c(2);
F=[];F(1)=A(1);
for i=2:(n+11)
    F(i)=(A(1)-b/a)/exp(a*(i-1))+b/a;
end
G=[];G(1)=A(1);
for i=2:(n+11)
    G(i)=F(i)-F(i-1);
end
t1=1976:2014;
t2=1976:2025;
t3=1977:2025;
y=diff(G);
plot(t1,A,'o',t2,G)
xlabel( 'years' )
ylabel( 'cases of infection ' )
grid on
plot(t1,A,'o',t3,y)
xlabel( 'years' )
ylabel( 'the infected people's rate of change ' )
grid on

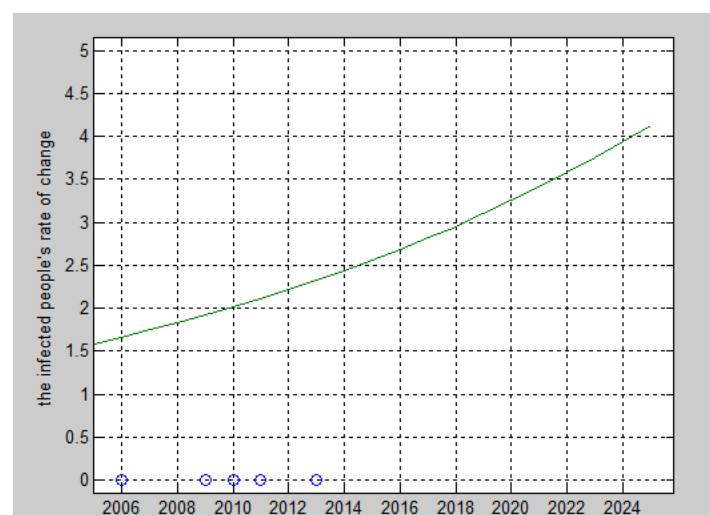
```



3-1 Picture shows the infected people's rate of change in Ebola virus.



3-2 The curve is cases of infected people in the future with infected by Ebola virus in Democratic Republic of Congo.



(the enlarged picture of(3-1) between 2006 and 2024)

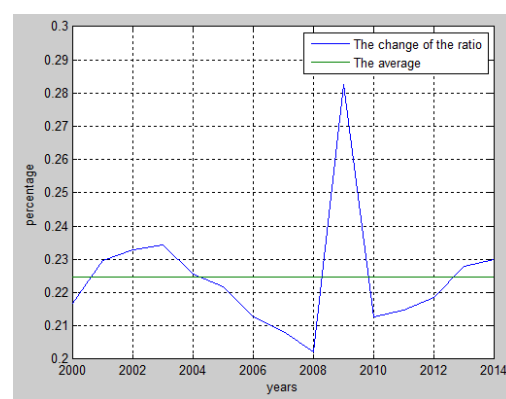
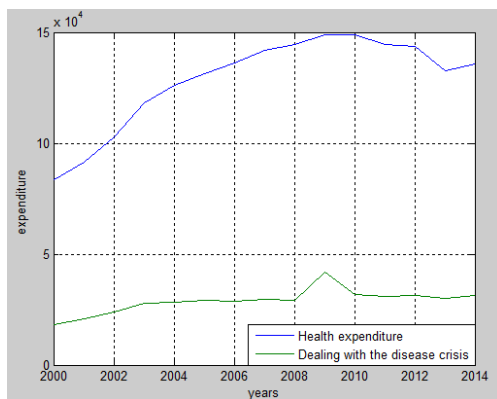
3-3 Picture shows the infected people people's rate of change in Ebola virus in 2006 to 2024.

Through the analysis of American economy, we figure out the finical assistance. The chart show the data of American to cope with the disease crisis over the years.

The data of American to cope with the disease crisis over the years (unit: million dollars)			
Years	The charge of health expenditure	The charge of dealing with the disease crisis	Percentage
2000	83581	18121	0. 216807648
2001	91505	21023	0. 229747008
2002	103057	23998	0. 232861426
2003	118065	27677	0. 234421717
2004	126270	28480	0. 225548428
2005	131259	29129	0. 221920021

2006	136019	28932	0. 212705578
2007	141890	29581	0. 208478399
2008	144391	29212	0. 202311778
2009	148739	42051	0. 282716705
2010	148962	31693	0. 212758959
2011	144379	30990	0. 214643404
2012	143737	31411	0. 218531067
2013	132477	30200	0. 227964099
2014	135665	31196	0. 229948771
Average	128666. 4	28912. 93333	0. 224757667

Through the economics of health from 2000 to 2014, we can get the curve by following. In the analysis of it, it's obvious that the increase financial in health. We can get it usually is about 128666.4 million dollars. And costs in dealing with disease crisis keep in 28912.93333million dollars.



In summary, we estimate the costs in America dealing with the crisis is about 32000 million dollars.

3.2 Logistic Model(we use this model into Task#2)

3.2.1 The Foundation of Logistic Model

Logistic regression model is analyzed in detail. By illustrating the relation between regression analysis and its probability hypothesis, analyzing and deducing regression models which are performed in the framework of generalized linear models, a thorough understanding of the theoretical background and construction methods is of benefit to the reasonable applications of these models.

3.3 Polynomial regression curve fitting

3.3.1 The Foundation of Polynomial regression curve fitting

Curve-fitting means we try to build some smooth curve, to get the data fitting better, and its not necessary for curve to cross the data point.

The basic form of a polynomial(we use this mathematic methods into Task#2)

(1)We can describe the basic form of a polynomial as,

$$f(x) = a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n$$

In the formula, the coefficient $a_0, a_1, \dots, a_n$ is the unknown coefficient need to be fitting.

(2)Polynomial fitting order:

Form:

<code>p=polyfit(x,y,n)</code>	%x,y are the same dimension incremental,n is polynomial fitting
<code>Y=polyval(p,x)</code>	P is fitting the coefficient of % %Y is the prediction data at x by getting polynomial fitting in polyfit
<code>[p,S]=polyfit(x,y,n)</code>	%S is the structure data, inclding R df and normr
<code>[Y,DELTA]=polyfit(x,y,S,alpha)</code>	the prediction data of % Y `s significant is 1-al- The confidence interval of %pha is $Y \pm DELTA$;
<code>[p,S,mu]=polyfit(x,y,n)</code>	By default time of %alpha is 0.5 %mu=[mean(x);std(x)],p is the fitting coefficient of centralized %
<code>[Y,DELTA]=polyconf(p,X,S,paraml,vall,param2,val2,...)</code>	The prediction of % named param `s choice is 'alpha', 'mu', % 'predopt' choice is 'alpha', 'mu', %'predopt', 'simopt', val is corresponding value.

We check the following informations then use them into the mathematic model and mathematic methods listed above.

3.3.2 Assumptions(in task#2)

(1)Different virus has different development speed. In our model, we focus on the virus which would be infected.

(2)Noninfected virus is not concerned in our model. When we confirm the reasonable of the study, we can completed within a reasonable period of time.

(3)The question we study about the population is according to the average population density.

3.3.3 The analysis of the problem(in task#2)

In the task#2, we study at the influence factor of drugs research. Finding the main influence resource is the virus development spread-x1(we can take it as the development number in infected people),rate of death-x2(the people dead contrast to the total population), deathrate of the change-y.

At first, we build the three curve including time line according to the chart. Assuming the three factor is connected. We deal with this question in equation of linear regression. Build the regression equation then through the data we can get the different coefficient and confirm the connection in three factors.

At last, we make use of the average in three influence factor, through the LOGISTIC Regression Model, then estimate the probability in drugs development. And we can get the curve about the probability with time line.

WHO report date	Total Cases, Liberia	Total Deaths, Liberia	infected	death	Speed of development-x1	Rate of death-x2	The rate of development in death-y	average-z
4/2014	8	2	157	76	8.8125	0.484076433	5.333333333	4.876636589
5/2014	13	11	61	49	-0.611464968	0.803278689	-0.355263158	-0.054483146
6/2014	13	9	140	96	1.295081967	0.685714286	0.959183673	0.979993309
7/2014	107	65	1665	945	10.89285714	0.567567568	8.84375	6.768058237
8/2014	391	227	8143	4389	3.890690691	0.538990544	3.644444444	2.691375226
9/2014	1871	1089	20875	11290	1.563551517	0.540838323	1.572339941	1.225576594
10/2014	3696	1998	46441	23771	1.224718563	0.511853755	1.105491585	0.947354633
11/2014	6525	2697	40995	17038	-0.11726707	0.415611669	-0.283244289	0.005033434

(we list the whole chart in Appendix)

3.3.4 Solution and Result(in task#2)

We wrote the following codes to build the curve with Logistic Model.

```

y=[5.3333 -0.3553 0.9592 8.8438 3.6444 1.5723 1.1055 -0.2832]';
x1=[8.8125 -0.6115 1.2951 10.8929 3.8907 1.5636 1.2247 -0.1173]';
x2=[0.4841 0.8033 0.6857 0.5676 0.5390 0.5408 0.5119 0.4156]';
X=[ones(8,1) x1,x2];
[b,bint,r,rint,stats]=regress(y,X)

```

%The result of these as follows:

```

b =
    -0.4696
     0.7394
     1.0225
bint =
    -4.1513     3.2122
     0.5666     0.9122
    -5.0016     7.0466
r =
    -1.2078
    -0.2549

```

-0.2299
 0.6792
 0.6862
 0.3329
 0.1462
 -0.1518

rint =

-1.7788 -0.6367
 -1.3909 0.8810
 -2.0036 1.5438
 -0.3110 1.6694
 -1.0555 2.4279
 -1.5318 2.1975
 -1.7019 1.9943
 -1.5425 1.2388

stats =

0.9623 63.7323 0.0003 0.5328

From the result, we can the obvious linear correlation with x1, x2 and y.

According to the Logistic Model we get the rate of drugs development and draw the curve:

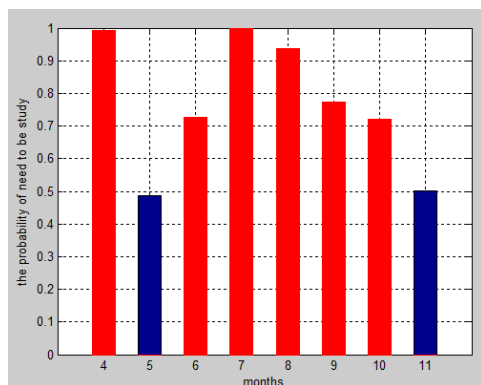
$$z = \frac{(y + x_1 + x_2)}{3} ;$$

$$p = 1 - \frac{1}{1 + e^z} ;$$

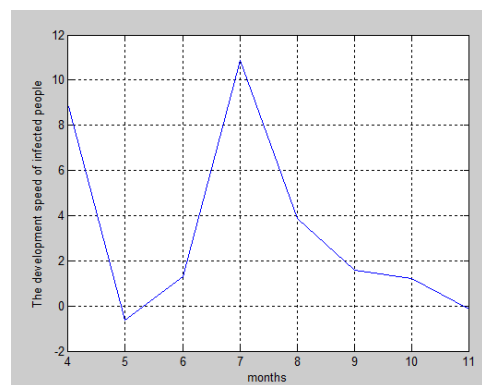
In line accordance with two formula above we get p following:

```
p=[0.9924 0.9924 0.4864 0.7271 0.9989 0.9365 0.7730 0.7206 0.5012]';
bar(p,0.5)
set(gca,'XTickLabel',{'4','5','6','7','8','9','10','11'})
xlabel('months'),ylabel('the probability of need to be study')
grid on
```

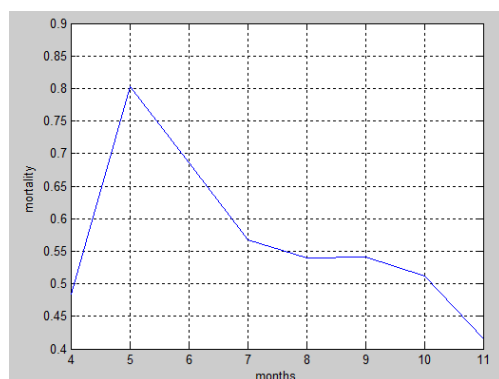
We use the data from the chart and mathematic methods above listed .



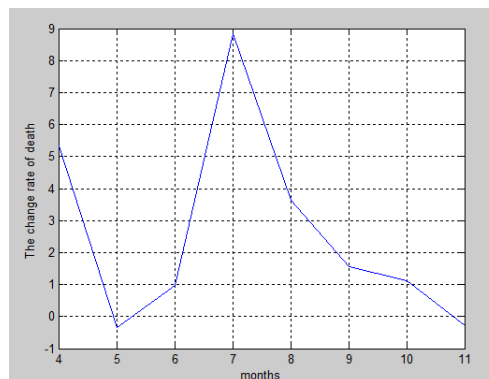
This curve is about p.



This curve is about x1.



This curve is about x2.



This curve is about y.

3.3.5 Analysis of the Result(in task#2)

Through the calculation, we can predict the probability of needing to study through the research about all of the original data of the disease. Now we analyzes probability in the case of Ebola. Through above probability histogram: we can get it clearly and directly that the research about Ebola virus last above 50%. It's imminent for us to study about the Ebola virus.

Through the curve to confirm that drugs need to be developed. It's deadline and the development speed is in inverse proportion function. We can assume $D_{\max} = \frac{\lambda T}{x_1}$.

The process of drugs development:

Extraction of virus → Experimental study → Determine the prescription → Animal experiment → Clinical trials → Batch production

3.4 On the basis budget prediction assumption models

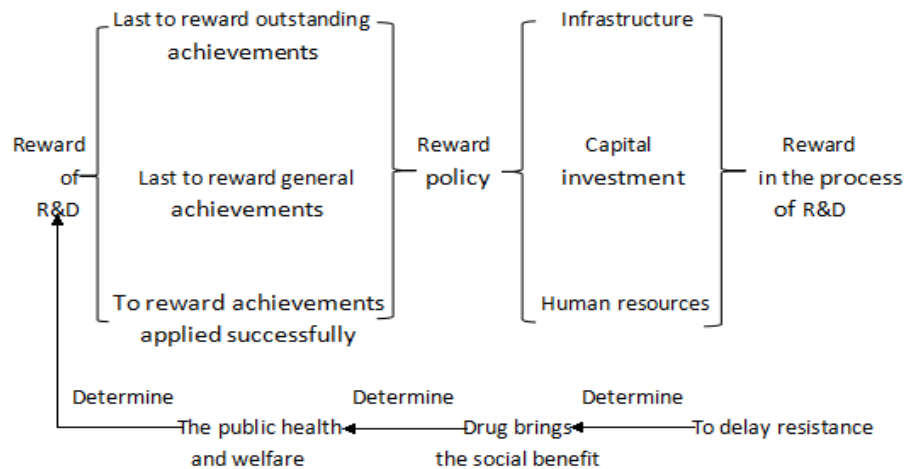
3.4.1 Assumptions(in task#3)

In task#3, we assume the financing in America's "reward" policy is in the budget prediction of USA. The result what we get are pragmatic value, means it is not useless.

3.4.2 Solution and Result(in task#3)

Focus on the “reward” policy, we depart it into two directions to study this task. One, “reward” in the drugs development .Two, “reward” in the development result.

Introduction:



①Public Health and Welfare: In this case, it mainly refers to the drugs development and delay the resistance to the drugs as much as possible. And the drug can respond to the crisis that common antibiotics can not solve.

②All reward is based with Bonus. If R & D achievements applied successfully, they will be awarded Honorary certificate.

③Assume:

All funds: W

Funds in the process of R&D: $W/3$

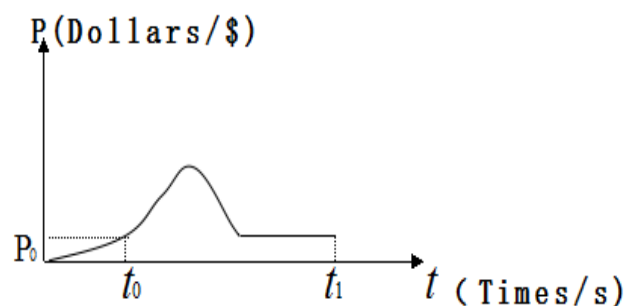
Funds of last to reward general: $W/6$

Funds of last to reward outstanding: $W/3$

Funds of to reward achievement applied successfully: $W/6$

④Graphic interpretation: The division standard reward of R&D achievements depend on Public Health and Welfare. The social benefit of drug is determined to Public Health and Welfare. And delaying drug of resistance determined social benefit of drug.

In qualitative analysis we can get the picture following:



P_0 :the maximum funds of general achievement

t_0 :the resistance to the drugs in the general achievement

t_1 :the resistance to the drugs in the outstanding achievement

3.4.3 Analysis of the Result(in task#3)

Based on the model we obtained reward policy

I .Reward policy in the process of R&D

- (1)To provide experimental equipment and experimental sites;
- (2)To offer Subsidies about Equipment loss fee of drug R&D, the experiment of raw materials and the chemical cost and so on.
- (3)To Encourage senior professor of colleges and universities to participate in the cooperative research project.

II .Reward policy of R & D achievements (more data visits above picture)

- (1) According to the resistance appeared in different time, we give the different degrees of reward to general achievements.
- (2)According to the resistance appeared in different time, we give the different degrees of reward to outstanding achievement.
- (3)With to reward achievement applied successfully, they will be awarded Honorary Certificate but bonus.

3.5 On the basis of task matching (allocation) models

3.5.1 The Foundation of task matching Model

In daily production, we always meet things like assign different people or different machine to accomplish the same workload and we know that one people or machine could accomplish one or more task. The task assignment models focus on the problem that how to assign reasonable for people (machine) work together be more effective.

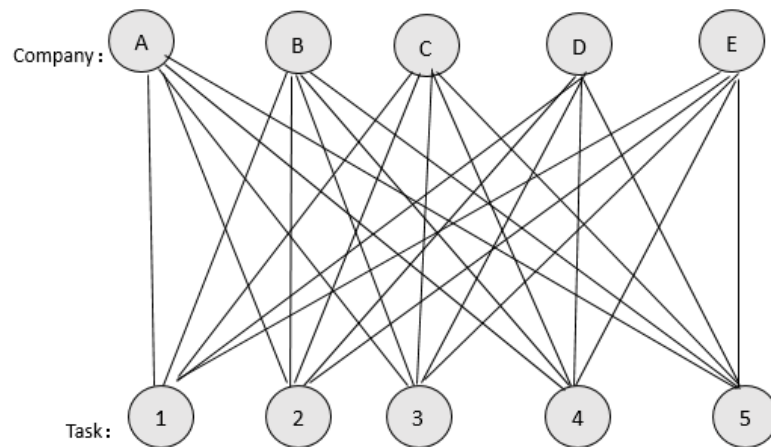
3.5.2 Assumptions(in task#4)

- (1)Each company only to complete a task.
- (2)Combination of every company and every task will have an associated cost and risk grade.
- (3)The pursuit of the goal is to determine how to make research and development costs, and the lowest risk.

3.5.3 Solution and Result(in task#4)

The assumed data list of each company for cost of tasks required (unit: Billion dollars)					
Company\mission	1	2	3	4	5
A	20	19	20	28	17
B	18	24	27	20	20
C	26	16	15	18	15
D	17	20	24	19	16
E	15	18	21	17	21
The of data list which risks of each company to complete the tasks assumed (additional: 1、2、3、4、5 show the grade is more serious)					
Company\mission	1	2	3	4	5
A	5	4	5	1	2

B	3	2	4	5	5
C	1	5	1	3	1
D	2	3	2	4	2
E	1	3	4	2	3



The picture of mission in assignment

In connection with data in the chart we build the program in MATLAB:

```

clc,clear
Number=input('please enter the codes of problem(1 min risks 2lowest cost) Number=: ');
if Number==1
    F=[5 4 5 1 2 3 2 4 5 5 1 5 1 3 1 2 3 2 4 2 1 3 4 2 3];
elseif Number==2
    F=[20 19 20 28 17 18 24 27 20 20 26 16 15 18 15 17 20 24 19 16 15 18 21 17 21];
end
A=[];B=[];
Aeq=[ones(1,5) zeros(1,20);zeros(1,5) ones(1,5) zeros(1,15);zeros(1,10) ones(1,5)
zeros(1,10);...
zeros(1,15) ones(1,5) zeros(1,5);zeros(1,20) ones(1,5);...
1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,4);...
zeros(1,1) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,3);...
zeros(1,2) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,2);...
zeros(1,3) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,1);...
zeros(1,4) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,4) 1 zeros(1,0)];
Beq=ones(1,10);
if Number==1
    [x,min_fval]=bintprog(F,A,B,Aeq,Beq);
    min_fval2=min_fval/5;
    fprintf('(1)the match function of company: \n')

```

```

    fprintf(' (A1) the match function of company A ( 1 replace match 0 replace not match ) :
\n')
    disp('    A-1    A-2    A-3    A-4    A-5'),disp(x(1:5))
    fprintf(' (B1) the match function of company B ( 1 replace match 0 replace not match ) :
\n')
    disp('    B-1    B-2    B-3    B-4    B-5'),disp(x(6:10))
    fprintf(' (C1) the match function of company C ( 1 replace match 0 replace not match ) :
\n')
    disp('    C-1    C-2    C-3    C-4    C-5'),disp(x(11:15))
    fprintf(' (D1) the match function of company D ( 1 replace match 0 replace not match ) :
\n')
    disp('    D-1    D-2    D-3    D-4    D-5'),disp(x(16:20))
    fprintf(' (E1) the match function of company E ( 1 replace match 0 replace not match ) :
\n')
    disp('    E-1    E-2    E-3    E-4    E-5'),disp(x(21:25))
    fprintf('min risks: '),disp(min_fval2)
elseif Number==2
    [x,min_fval]=bintprog(F,A,B,Aeq,Beq);
    min_fval2=min_fval/5;
    fprintf('(2)the match function of company: \n')
    fprintf(' (A2) the match function of company A ( 1 replace match 0 replace not match ) :
\n')
    disp('    A-1    A-2    A-3    A-4    A-5'),disp(x(1:5))
    fprintf(' (B2) the match function of company B ( 1 replace match 0 replace not match ) :
\n')
    disp('    B-1    B-2    B-3    B-4    B-5'),disp(x(6:10))
    fprintf(' (C2) the match function of company C ( 1 replace match 0 replace not match ) :
\n')
    disp('    C-1    C-2    C-3    C-4    C-5'),disp(x(11:15))
    fprintf(' (D2) the match function of company D ( 1 replace match 0 replace not match ) :
\n')
    disp('    D-1    D-2    D-3    D-4    D-5'),disp(x(16:20))
    fprintf(' (E2) the match function of company E ( 1 replace match 0 replace not match ) :
\n')
    disp('    E-1    E-2    E-3    E-4    E-5'),disp(x(21:25))
    fprintf('lowest costs: '),disp(min_fval2)
end

```

%The result of these as follows:

Please enter the codes of problem (1min risks 2 lowest cost) Number=: 1

(1) the match function of company:

(A1) the match function of company A (1 replace match 0 replace not match) :

A-1 A-2 A-3 A-4 A-5

0 0 0 1 0

(B1) the match function of company B (1 replace match 0 replace not match) :

B-1	B-2	B-3	B-4	B-5
0	1	0	0	0

(C1) the match function of company C (1 replace match 0 replace not match) :

C-1	C-2	C-3	C-4	C-5
0	0	0	0	1

(D1) the match function of company D (1 replace match 0 replace not match) :

D-1	D-2	D-3	D-4	D-5
0	0	1	0	0

(E1) the match function of company E (1 replace match 0 replace not match) :

E-1	E-2	E-3	E-4	E-5
1	0	0	0	0

min risks: 1.4000

Please enter the codes of problem (1min risks 2 lowest cost) Number=: 2

(1) the match function of company:

(A2) the match function of company A (1 replace match 0 replace not match) :

A-1	A-2	A-3	A-4	A-5
0	1	0	0	0

(B2) the match function of company B (1 replace match 0 replace not match) :

B-1	B-2	B-3	B-4	B-5
0	0	0	1	0

(C2) the match function of company C (1 replace match 0 replace not match) :

C-1	C-2	C-3	C-4	C-5
0	0	1	0	0

(D2) the match function of company D (1 replace match 0 replace not match) :

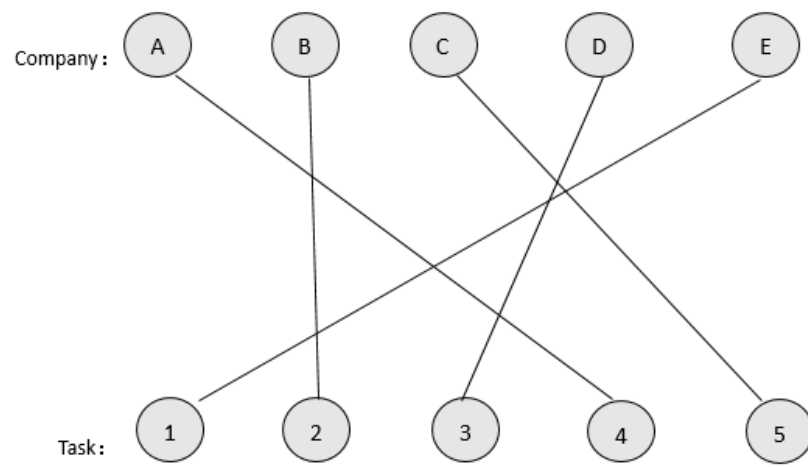
D-1	D-2	D-3	D-4	D-5
0	0	0	0	1

(E2) the match function of company E (1 replace match 0 replace not match) :

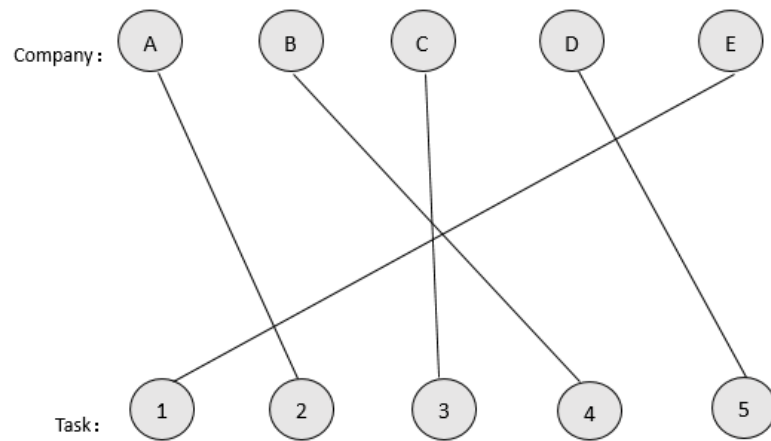
E-1	E-2	E-3	E-4	E-5
1	0	0	0	0

lowest costs: 17

From the codes we write above we can draw the following curve:



The picture of mission in assignment(min risks)



The picture of mission in assignment(lowest cost)

IV. References

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V. Appendix

5.1 chart(used in the task#2)

WHO report date	Total Cases	Total Deaths	infected	death	Speed of development-x1	Rate of death-x2	The rate of development in death-y	average-z
3/1/2014	0	0	16	12				
3/2/2014	0	0						
3/3/2014	0	0						
3/4/2014	0	0						
3/5/2014	0	0						
3/6/2014	0	0						
3/7/2014	0	0						
3/8/2014	0	0						
3/9/2014	0	0						
3/10/2014	0	0						
3/11/2014	0	0						
3/12/2014	0	0						
3/13/2014	0	0						
3/14/2014	0	0						
3/15/2014	0	0						
3/16/2014	0	0						
3/17/2014	0	0						
3/18/2014	0	0						
3/19/2014	0	0						
3/20/2014	0	0						
3/21/2014	0	0						
3/22/2014	0	0						
3/23/2014	0	0						
3/24/2014	0	0						
3/25/2014	0	0						
3/26/2014	0	0						
3/27/2014	8	6						
3/31/2014	8	6						
4/1/2014	8	2	157	76	8.8125	0.484076433	5.33333333	4.876636589
4/2/2014	8	5						
4/7/2014	18	7						
4/10/2014	22	14						

4/17/2014	27	13						
4/21/2014	27	13						
4/23/2014	34	11						
4/30/2014	13	11						
5/5/2014	13	11	61	49	-0.611464 968	0.8032786 89	-0.355263 158	-0.054483 146
5/14/2014	12	11						
5/23/2014	12	9						
5/27/2014	12	9						
5/28/2014	12	9						
6/2/2014	13	9	140	96	1.2950819 67	0.6857142 86	0.9591836 73	0.9799933 09
6/5/2014	13	9						
6/10/2014	15	10						
6/11/2014	15	10						
6/18/2014	33	24						
6/24/2014	51	34						
7/2/2014	107	65	1665	945	10.892857 14	0.5675675 68	8.84375	6.7680582 37
7/7/2014	115	75						
7/8/2014	131	84						
7/14/2014	142	88						
7/16/2014	172	105						
7/21/2014	196	116						
7/24/2014	224	127						
7/28/2014	249	129						
7/31/2014	329	156						
8/3/2014	391	227	8143	4389	3.8906906 91	0.5389905 44	3.6444444 44	2.6913752 26
8/4/2014	486	255						
8/6/2014	391	227						
8/8/2014	554	294						
8/12/2014	599	323						
8/13/2014	670	355						
8/15/2014	786	348						
8/19/2014	834	466						
8/21/2014	972	576						
8/22/2014	1082	624						
8/28/2014	1378	694						
9/6/2014	1871	1089	2087 5	1129 0	1.5635515 17	0.5408383 23	1.5723399 41	1.2255765 94
9/8/2014	2046	1224						
9/12/2014	2081	1137						

9/16/2014	2407	1296						
9/18/2014	2710	1459						
9/22/2014	3022	1578						
9/24/2014	3280	1677						
9/26/2014	3458	1830						
10/1/2014	3696	1998	4644 1	2377 1	1.2247185 63	0.5118537 5	1.1054915 85	0.9473546 33
10/3/2014	3834	2069						
10/8/2014	3924	2210						
10/10/2014	4076	2316						
10/15/2014	4249	2458						
10/17/2014	4262	2484						
10/22/2014	4665	2705						
10/25/2014	4665	2705						
10/29/2014	6535	2413						
10/31/2014	6535	2413						
11/5/2014	6525	2697	4099 5	1703 8	-0.117267 07	0.4156116 6	-0.283244 289	0.0050334 34
11/7/2014	6619	2766						
11/12/2014	6822	2836						
11/14/2014	6878	2812						
11/19/2014	7069	2964						
11/21/2014	7082	2963						